

Cloud Integration & OpenStack

Formation · Lesson 13 (Capstone)

Built from the official documentation

`kubernetes-sigs/kubespray` · cloud_controllers · CSI

Learning Objectives

By the end of this lesson you will be able to:

- Explain the **external cloud-provider** model and why in-tree is gone
- Enable OpenStack with `cloud_provider: external` + `external_cloud_provider`
- Supply **OpenStack credentials** (openstack-rc or application credentials)
- Configure **Octavia/LBaaS** so Services get a floating IP
- Deploy **Cinder CSI** and a StorageClass for dynamic volumes
- **Verify** the integration: pods, a PVC, a LoadBalancer

Capstone: wire a Kubespray cluster into a real cloud — primarily OpenStack.

22.1 The External Cloud Provider Model

In-Tree Is Removed — Use the External CCM

Since Kubernetes **v1.31** (Kubespray **v2.27**), Kubernetes no longer supports `cloud_provider` **except the external cloud provider**.

Concept	In-tree (removed)	External (current)
CCM	inside controller-manager	separate <code>cloud-controller-manager</code> pod
Enable	<code>cloud_provider: openstack</code>	<code>external</code> + <code>external_cloud_provider</code>
Storage	in-tree volume plugin	Cinder CSI (<code>cinder.csi.openstack.org</code>)
LBaaS	Neutron LBaaS	Octavia (default)

Storage moved to **out-of-tree CSI**; the CCM and Cinder CSI are configured separately and can even run independently.

22.2 Enabling OpenStack

Turn On the External OpenStack Provider

Set two variables in `group_vars/all/all.yml`, then re-run `cluster.yml`:

```
cloud_provider: external
external_cloud_provider: openstack
```

- Kubespray configures the provider **to use Octavia by default** (not the old Neutron LBaaS).
- The same playbook deploys the **cloud-controller-manager** and (when enabled) **Cinder CSI** pods.
- vSphere uses the identical pattern with `external_cloud_provider: vsphere`.

Inventory **hostnames must match** the OpenStack instance names — otherwise Cinder will not work as expected.

OpenStack Credentials

Source your `openstack-rc` before running the playbook:

```
# exports OS_AUTH_URL, OS_USERNAME, OS_PASSWORD, ...  
source path/to/your/openstack-rc
```

Or use Keystone **application credentials** (precedence over username/password; **all three required**) in `group_vars/all/openstack.yml`:

```
external_openstack_application_credential_name:  
external_openstack_application_credential_id:  
external_openstack_application_credential_secret:
```

Multi-NIC VMs: override `external_openstack_network_internal_networks` / `_public_networks` and `external_openstack_metadata_search_order`.

22.3 Octavia / LBaaS

LBaaS Variables (group_vars/all/openstack.yml)

```
external_openstack_lbaas_enabled: true
external_openstack_lbaas_floating_network_id: "Neutron network ID for floating IP"
external_openstack_lbaas_subnet_id: "Neutron subnet ID to create LBaaS VIP"
external_openstack_lbaas_provider: amphora          # or ovn
external_openstack_lbaas_method: ROUND_ROBIN
external_openstack_lbaas_manage_security_groups: false
external_openstack_lbaas_create_monitor: false
```

- `..._floating_network_id` gives a `LoadBalancer` Service its **floating IP**.
- `..._provider` selects the Octavia driver: **amphora** (default) or **ovn**.

Migrating? Replace the old `openstack_lbaas_subnet_id` with the new `external_openstack_lbaas_subnet_id` — the new provider uses **Octavia**.

22.4 Cinder CSI

Enable Cinder CSI

In `group_vars/all/openstack.yml` (after sourcing openstack-rc):

```
cinder_csi_enabled: true
cinder_csi_controller_replicas: 1
cinder_csi_ignore_volume_az: true    # schedule on any node AZ
cinder_topology: true                # optional topology support
```

- Cinder CSI does **not** require enabling the OpenStack CCM — both can coexist (their provisioners are named differently).
- To also deploy the provisioner, set `persistent_volumes_enabled: true` in `group_vars/k8s_cluster/k8s_cluster.yml`.
- Snapshotter & sidecars are pinned via `cinder_csi_snapshotter_image_tag`, `..._attacher`, `..._provisioner`, `..._resizer`, `livenessprobe`.

Only **Cinder v3** is supported by the CSI driver.

Cinder StorageClass

```
storage_classes:
- name: "cinder-csi"
  provisioner: "cinder.csi.openstack.org"
  mount_options:
  - "discard"
  parameters:
    type: "thin"
    availability: "nova"
    reclaim_policy: "Delete"
    volume_binding_mode: "WaitForFirstConsumer"
```

- Default class is `cinder-csi`, real provisioner `cinder.csi.openstack.org`.
- `WaitForFirstConsumer` binds the volume only once a Pod is scheduled.

22.5 Other Clouds

vSphere / AWS / Azure / GCP

vSphere — same model, in `group_vars/all/vsphere.yml`:

```
cloud_provider: "external"
external_cloud_provider: "vsphere"
external_vsphere_vcenter_ip: "myvcenter.domain.com"
external_vsphere_vcenter_port: "443"
external_vsphere_insecure: "true"
external_vsphere_user: "administrator@vsphere.local"
external_vsphere_password: "K8s_admin"
external_vsphere_datacenter: "DATACENTER_name"
vsphere_csi_enabled: true
```

- Requires vSphere **6.7 U3+**, VM hardware **v15+**, **UUID activated** per VM.
- **AWS / GCP** legacy flag `-e cloud_provider=[aws|gce]`; in-tree removed
≥ v1.31 → use the external CCM + matching CSI per cloud (Azure, OCI, Huawei).

22.6 Verification

Check Pods, PVC and LoadBalancer

```
# cloud-controller + cinder-csi pods Running
kubectl -n kube-system get pods | grep cinder
# csi-cinder-controllerplugin-... 5/5 Running
# csi-cinder-nodeplugin-... 2/2 Running

kubectl get storageclass      # cinder-csi cinder.csi.openstack.org

# apply nginx.yaml (PVC + Pod) → volume provisioned & bound
kubectl get pvc              # csi-pvc-cinderplugin Bound 1Gi cinder-csi
kubectl exec -it nginx -- df -h | grep /var/lib/www/html # /dev/vdb mounted

kubectl get svc              # LoadBalancer EXTERNAL-IP = Octavia floating IP
```

Result: a **PVC provisions a Cinder volume** and a **LoadBalancer Service gets a floating IP** — the integration is working end to end.

L3 CNl on OpenStack (calico / kube-router)

OpenStack drops packets from IPs it does not know (anti-spoofing). L3 CNIs route with **pod IPs directly**, so you must allow them on the Neutron ports:

```
openstack port set <PORT_ID> \  
  --allowed-address ip-address=10.233.0.0/18 \  
  --allowed-address ip-address=10.233.64.0/18
```

- `10.233.0.0/18` = `kube_service_addresses`, `10.233.64.0/18` = `kube_pods_subnet`.
- Set on **every** instance port (sweep with `openstack port list | xargs`).

Skip this and pod traffic silently fails on calico / kube-router.

Key Takeaways

- **External CCM + CSI** is the only supported model: `cloud_provider: external`
 - `external_cloud_provider: openstack`.
- Auth = **source openstack-rc** or **application credentials** (all three);
inventory hostnames must match instance names.
- **Cinder CSI** (`cinder_csi_enabled: true`) → dynamic volumes; **Octavia** → `LoadBalancer` Services with a floating IP.
- **vSphere / AWS / Azure / GCP** follow the same external-CCM + CSI pattern.

Formation Complete

You can now deploy & operate Kubespray clusters — including on OpenStack

`cloud_provider: external` · `cinder_csi_enabled` · Octavia LoadBalancers